

Estimating Sampling Rate of Human Activity Data from Accelerometer using Transformer-based Regression Model

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ABSTRACT

We propose a method to estimate the sampling frequency from only the 3-axis acceleration data. The proposed method calculates the absolute difference between two consecutive samples, creates a histogram, and constructs a Transformer-based regression model.

CCS CONCEPTS

• Human-centered computing → Ubiquitous computing.

KEYWORDS

acceleration data, sampling frequency, regression model, transformer

ACM Reference Format:

Hinase Kawano, Marina Okamoto, and Kazuya Murao. 2023. Estimating Sampling Rate of Human Activity Data from Accelerometer using Transformer-based Regression Model. In *Adjunct Proceedings of the 2023 ACM International Joint Conference on Pervasive and Ubiquitous Computing & the 2023 ACM International Symposium on Wearable Computing (UbiComp/ISWC '23 Adjunct)*, October 08–12, 2023, Cancun, Quintana Roo, Mexico. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3594739.3610794>

1 INTRODUCTION

It is important to know the sampling frequency of the acceleration data to be processed. Incorrect or missing frequency information can lead to improper system operation. We propose a method to estimate the sampling frequency from only the acceleration data.

2 PROPOSED METHOD

The average of long acceleration data $m = \sqrt{m_x^2 + m_y^2 + m_z^2}$ can be approximated by the acceleration of gravity. For the interval where the 3-axis resultant acceleration a_k satisfies $0.9m < a_k < 1.1m$ for five or more consecutive samples, the user is considered stationary because the acceleration is close to gravity, and the data in that interval is eliminated. As the sampling frequency increases, the time interval between data recordings becomes shorter, causing the acceleration value sensed at one time to become smaller. Therefore, the differential values of the resultant acceleration are considered to have unique characteristics for each sampling frequency. The absolute value of the difference between two consecutive samples

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UbiComp/ISWC '23 Adjunct, October 08–12, 2023, Cancun, Quintana Roo, Mexico
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ACM ISBN 979-8-4007-0200-6/23/10.
<https://doi.org/10.1145/3594739.3610794>

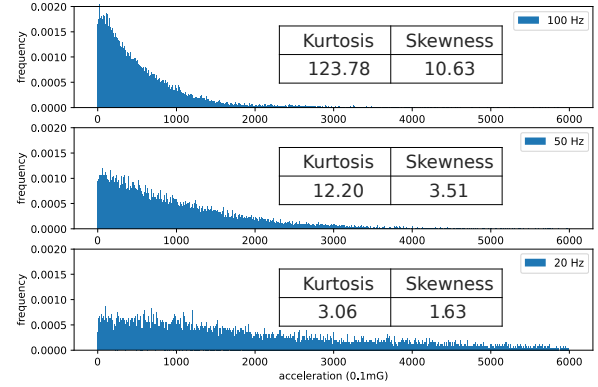


Figure 1: Histograms of difference value at 100, 50, 20 Hz.

Table 1: Result of sampling frequency estimation.

Data domain (train→test)	MAE	RMSE	R^2	SD
Our data → Our data	1.311	1.732	0.995	2.582
HASC2014 → HASC2014	0.744	1.202	0.998	1.677
Our data → HASC2014	9.547	12.168	0.784	12.078
HASC2014 → Our data	9.617	12.583	0.466	11.630

is calculated by $d_j = |a_{j+1} - a_j|$ from the resultant acceleration a_j , where $j = 1, \dots, N - 1$ is time. As shown in Fig. 1, a histogram with 4,000 bins is created from d_j of each sampling frequency. We build a regression model using Transformer, where the explanatory variable is a histogram with 4,000 bins and the objective variable is a 1-dimensional sampling frequency.

3 EVALUATION EXPERIMENT

An evaluation experiment was conducted using 225 data samples prepared by one of the authors and 468 data samples from the HASC Corpus 2014 [1]. The results are shown in Table 1. Errors between the same data sets were approximately 1 Hz, while the errors between different data sets were approximately 10 Hz.

4 CONCLUSION

We proposed the sampling frequency estimation method for acceleration data and confirmed that the estimation error is 1 Hz. In the future, we use more data sets including diverse activities to improve the accuracy and apply our method to other sensors.

REFERENCES

- [1] Kawaguchi et al. 2011. HASC Challenge: gathering large scale human activity corpus for the real-world activity understandings. In *AH 2011*. 1–5.